

# Deva Anand

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## Education

### University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

**Coursework:** Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

### Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

## Skills

### Languages

JavaScript, TypeScript, Python, Java, C++, SQL, HTML/CSS, Bash

### Frontend & Web

React, Redux, Next.js, REST APIs, HTTP, responsive UI, Electron, Webpack

### Backend & Data

Node.js, Express, FastAPI, Fastify, async services, data modeling, auth, MongoDB, PostgreSQL, Redis

### Serverless & Edge

Cloudflare Workers AI, Durable Objects, WorkflowEntrypoint, D1, WebSockets, Docker, AWS, GCP, Azure

### AI-Assisted Dev & Agents

Cursor, Claude / Claude Code, GitHub Copilot, ChatGPT / Codex, LLM agents, MCP / APIs, Git, GitHub Actions, CI/CD

## Experience

### Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Engineered **multi-tenant NHS eTriage platform** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions** as a production-grade system integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Built a full-stack **internal tool** from scratch (**React, Node.js, MongoDB**), a data annotation platform used daily by the AI team, replacing manual spreadsheet workflows with secure role-based REST APIs; cut manual operations by **100+ hours/month** and improved throughput by **60%**, iterating directly with the AI-team users who relied on it daily.
- Owned backend reliability across the eTriage stack: monitored production logs during feature releases and client integrations, shipped fixes for tenant-specific incidents, hardened authentication and rate limiting, and remediated **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings** to protect sensitive clinical data.

### Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and prototyped LLM-powered context-aware replies in a chat product, improving type safety, maintainability, and end-user response quality.

## Projects

### cf-ai-research-scout - Edge-Native AI Agent on Cloudflare

github.com/Deva-1903/cf\_ai\_research\_scout

TypeScript, Cloudflare Workers AI, AIChatAgent, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built an **edge-native AI agent** end-to-end on **Cloudflare**: it fetches web sources, reasons through multi-step research, and streams responses to the client over **WebSocket** in real time, deployed entirely on Cloudflare Workers with no external backend.
- Designed durable workflow boundaries via **WorkflowEntrypoint** and **Durable Objects** so multi-step tasks survive process restarts and tool failures, with agent state and conversation history persisted server-side in **D1**.

### AgenticSearch - Provenance-First AI Agent

github.com/Deva-1903/ciir\_agentic\_search

Python, FastAPI, OpenAI / Groq, SQLite, Docker

- Developed a multi-stage **AI agent** that plans typed retrieval, reranks pages, extracts entities with deterministic parsers before LLM fallback, verifies outputs, and returns cell-level provenance for every result.
- Hardened the service with **150+ automated tests**, provider routing/fallback across LLM backends, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency.

### Underdogs Fitness - Production Full-Stack Web App

underdogsfitness.in

React + Redux, Node.js, Express, MongoDB, Stripe | solo, live in production 3+ years

- Built and deployed a **production gym-management web app** solo end-to-end (MERN): auth, membership and attendance management, Stripe + manual-cash payments with a partial-payment ledger, and cron-based renewal automation, live and in use for **3+ years**.
- Modeled the data for **multi-branch** from day one (every entity scoped by `branchId`, JWT-claim auth, branch-context frontend); when the client opened a 2nd branch a year later, onboarded it as a config change with **zero schema migration**.

## Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.